

Optimization

Pre-requisite

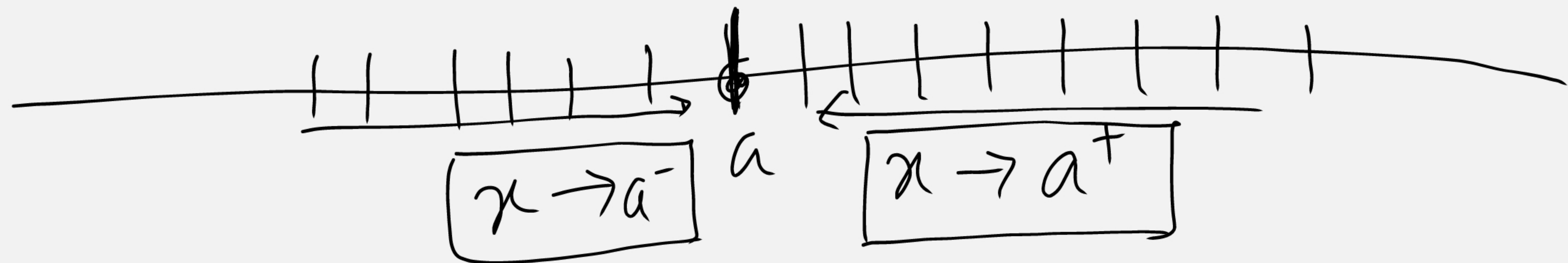
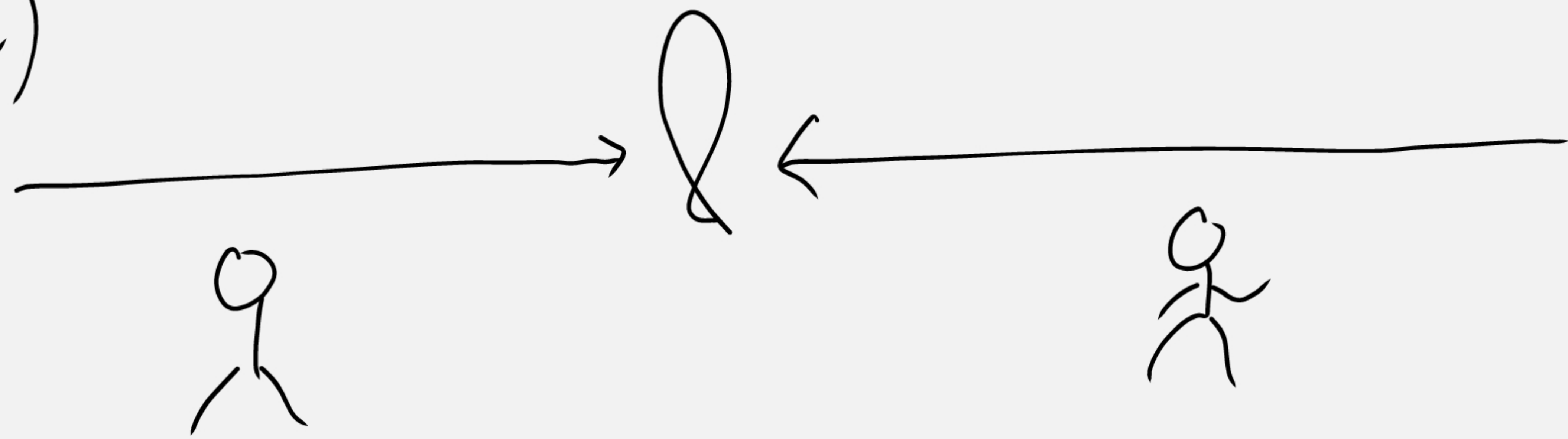
- 1) Limit
 - 2) Derivatives
 - 3) Continuity
 - 4) Maxima & Minima
- 

Topics covered

- 1) LA ✓
- 2) Probability & Statistics ✓
- 3) Discrete Mathematics ✓
- 4) Optimization.

Limit

$$y = f(x)$$



$$x \rightarrow a^-$$

L.H.L

$$x \rightarrow a^+$$

R.H.L

Left Hand Limit
(LHL)

$$\lim_{x \rightarrow a^-} f(x) = \text{exists \& finite}$$

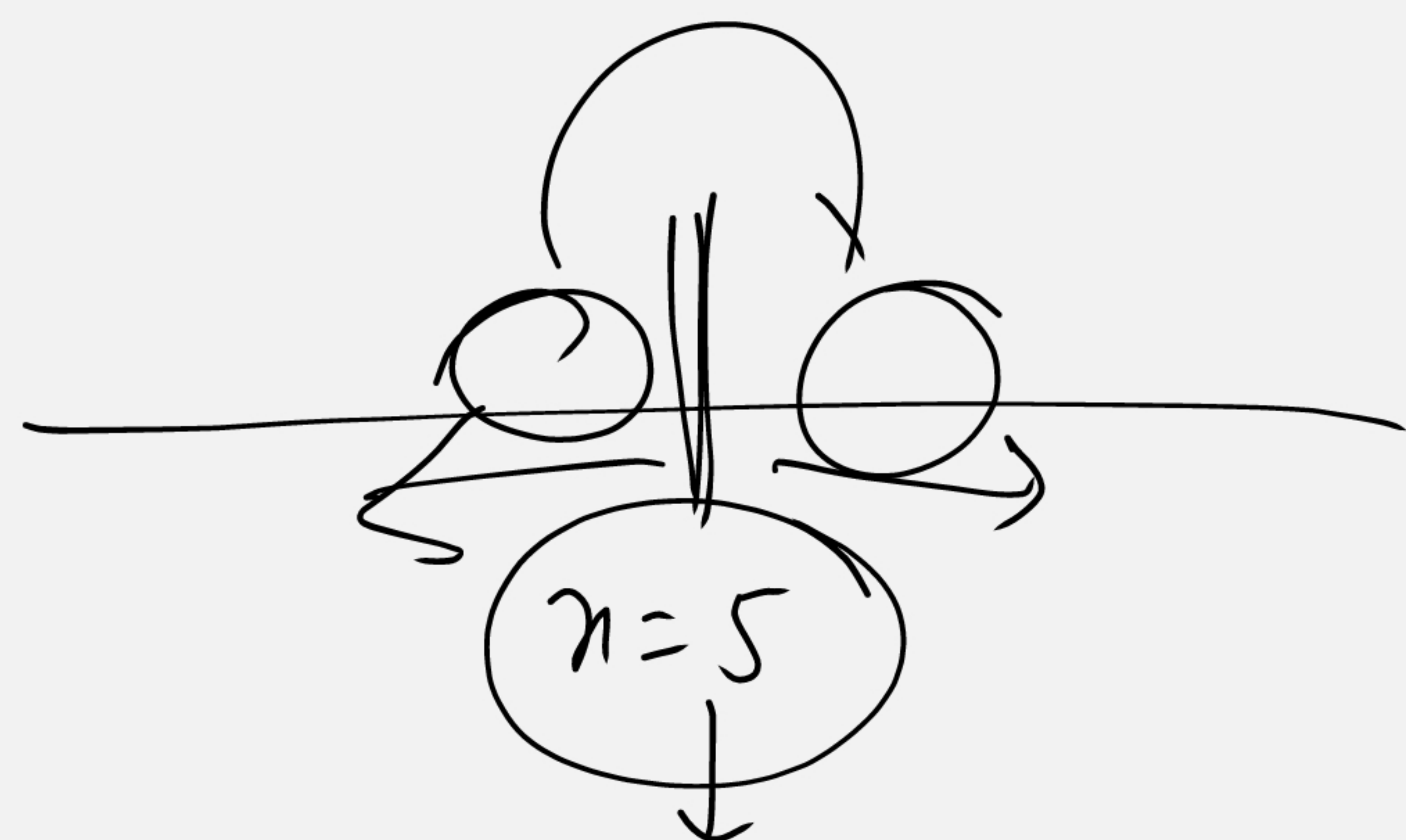
Right hand
limit
(RHL)

$$\lim_{x \rightarrow a^+} f(x) = \text{exists \& finite}$$

$$\lim_{x \rightarrow a} f(x) = L$$

Q. Discuss the limit of the function $f(x) = x + 1$ at $x = 0$

x					-0.1	-0.001	-0.0001	0.0001	0.001	0.01	0.1		
$f(x)$							0.9999	1.0001					



$x \rightarrow 0^-$
 $x \rightarrow 0^+$

$\lim_{x \rightarrow 0} f(x) = 1$

Q.1

$$\lim_{n \rightarrow 3} \frac{n^2 + 4}{n - 2}$$

Q.2

$$\lim_{x \rightarrow 1} \frac{2x + 1}{3x^2 - 4x + 5}$$

Q.

Evaluate the left-hand limit and right-hand limits of the func $x \neq 1$, if $0 < x < 1$

$$f(x) = \begin{cases} 1 + x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } x > 1 \end{cases}$$

does $\lim_{x \rightarrow 1} f(x)$ exist?

$$f(x) = 1 + x^2$$

$$x \rightarrow 1$$

$$\Rightarrow 1 + (1)^2$$

$$= 1 + 1$$

$$f(x) = 2 \quad (0 < x < 1)$$

$$\begin{cases} f(x) = 2 - x \\ x \rightarrow 1 = 2 - 1 \end{cases}$$

$$1 \neq 2$$

$$L.H.L \neq R.H.L$$

$$x = 1$$

$$x \rightarrow 1^-$$

$$x \rightarrow 1^+$$

find,
 $\lim_{x \rightarrow 10} (4x + 3\sqrt{x-2})$

$$= 4 \lim_{x \rightarrow 10} (x) + \lim_{x \rightarrow 10} (x-2)^{1/3}$$

$$= 4(10) + (10-2)^{1/3}$$

$$40 + 2 = 42 \quad \checkmark \checkmark$$

//

Indeterminate forms of limit-

$$\left(\frac{0}{0}\right), \left(\frac{\infty}{\infty}\right), \left(\infty - \infty\right), \left(\infty \times 0\right),$$

$0^0, \infty^0, 1^\infty$

first derivative

first-

Q.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{0}{0}$$

Indeterminate

Q.

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \infty$$

Q.

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x^2 - 3x + 2} \right) \frac{(x+2)}{(x-2)(x-1)} = \frac{0}{0}$$

$\lim_{x \rightarrow 2} \frac{(x+2)}{(x-1)}$

$\frac{4}{1} = 4$

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$2) \lim_{x \rightarrow 0} \cos x = 1$$

$$3) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$4) \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

$$5) \lim_{x \rightarrow 0} (1+nx)^{\frac{1}{x}} = e^n$$

$$6) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$7) \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = e^a$$

expansion

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Derivative

$$1) (f+g)' = f' + g'$$

$$2) (f-g)' = f' - g'$$

$$3) (f \cdot g)' = f g' + g f'$$

$$4) \left(\frac{f}{g}\right)' = \frac{g f' - f \cdot g'}{g^2}$$

$f(x)$ $f'(x)$ $f(x)$ $f'(x)$ x^n nx^{n-1} $\cos x$ $-\sin x$ $\tan x$ $\sec^2 x$ $\ln x$ $\frac{1}{x}$ $\sec x$ $\sec x \tan x$ e^x e^x $\csc x$ $-\csc x \cot x$ a^x $a^x \log_e a$ $\cot x$ $-\csc^2 x$ $\sin x$ $\cos x$ $\sinh x$ $\cosh x$

Hyperbolic

Maxima
&
Minima

Q.

$f(x) = 2x^3 - 3x^2 - 36x + 2$ has its maxima at

- (A) $x = -2$ only
- (B) $x = 0$ only
- (C) $x = 3$ only

- (D) both $x = -2$ and $x = 3$

Optimization
↓
Optimal soln. → Model
↓
Neural.

Procedure

$$f'(x) = 0$$

$$f''(x) = \begin{cases} > 0 \text{ (minima)} \\ < 0 \text{ (maxima)} \end{cases}$$



Parallel to x -axis
tangent / slope

$$f(x) = 2x^3 - 3x^2 - 36x + 2$$

$$f'(x) = 6x^2 - 6x - 36$$

$$\text{let } f'(x) = 0$$

$$6x^2 - 6x - 36 = 0$$

$$6(x^2 - x - 6) = 0$$

$$x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$x(x-3) + 2(x-3) = 0$$

$$\boxed{x = 3, -2}$$

$$f''(x) = 12x - 6$$

$$\text{When } x=3, 12(3) - 6 = \cancel{30} \text{ } 30 \\ = +ve (\text{Minima})$$

$$\text{When } x=-2, 12(-2) - 6 = -\cancel{30} \text{ } 30 \\ = -ve (\text{Maxima})$$

Consider the
function
 $f(x) = (x^2 - 4)^2$
where x is a real no.
Then the function has.

- (a) Only one minimum
- (b) Only two minima
- (c) Three minima
- (d) Three maxima